The ϵ of Imputation

Look-Ahead Leakage in Time-Series Imputation is Measurable, Pervasive, and Unnecessary — A Position and a Standard

Derek Snow[1], Matthew Lyberg[1], and Eeshaan Asodekar[1]

[1]Sovai Research d.snow@sov.ai

Abstract

The time-series imputation field benchmarks almost exclusively in the *bidirectional* (smoothing) setting: a method may use the future to fill the past. For any sequential use of the filled data—a backtest, an online controller, an early-warning monitor, training a downstream forecaster—this is look-ahead leakage, and it silently inflates the published score above anything achievable live. We argue three things. **(1) It is measurable.** We promote a model-agnostic look-ahead audit to a proposed community standard: wrap any imputer $f(X) \to \hat{X}$ and emit two numbers—a binary *causality certificate* (the maximum revision of an early imputed cell as the series grows, i.e. truncation invariance) and a continuous *leakage score* $\Delta = \text{MAE}_{\text{causal}} - \text{MAE}_{\text{bidir}}$, the accuracy a method borrows from the future. We call it the ϵ of imputation: a single mechanical test that turns "is this method causal?" from an argument into a measurement. **(2) It is pervasive.** Running the audit over 28 methods × 16 real tasks, the certificate cleanly partitions the field: 18/28 methods certify with zero revision and $\Delta \approx 0$, while 10/28 leak—interpolation, batch low-rank, and, revealingly, the three deep imputers we audit (SAITS, BRITS, a Transformer), whose bidirectional reconstruction is *beaten by their own honest right-edge filter* ($\Delta = -0.30$ to -0.38). A closed form $\Delta(a, g)$, validated against exact Kalman-filter/RTS-smoother runs at $R^2 = 0.9996$, predicts exactly which panels the leak inflates (high lag-1 memory) and which give the future nothing. We corroborate with documented field incidents where leakage forced retractions and re-runs. **(3) It is unnecessary.** CAFÉ, a causal, CPU-only statistical imputer, is a constructive existence proof: it certifies at $\Delta = 0$ and posts the lowest causal MAE in our survey (0.286 vs. 0.301 for the best causal rival). We recommend the field adopt the certificate-and-Δ as a reporting norm alongside MAE. This is a standards contribution, not a method-beats-SOTA claim.

1 Position

Missing data is the rule in multivariate time series and panels: sensors drop out, markets close, filings arrive irregularly, panels are unbalanced. The dominant evaluation protocol holds out a set of cells, fills them with the method under test applied to the *whole* matrix, and reports the error on the held-out cells. That protocol is *bidirectional*: the fill at time t is free to read observations at times $> t$. For a great many downstream uses of imputed data this is invalid. A backtest that trades on a feature whose value was reconstructed using future prices is not a backtest; an online monitor that "detects" an event using data from after the event is not a monitor; a forecaster trained on bidirectionally imputed features inherits a look-ahead it can never reproduce in deployment. The leak is not exotic—it is the default.

The position. *Look-ahead leakage in imputation is measurable, pervasive, and unnecessary, and the field should report a causality certificate and a leakage score alongside MAE.* We are not proposing a new imputer as the centrepiece. We are proposing (i) a measurement, (ii) a survey establishing that the leak is endemic to the methods that win the standard leaderboards, and (iii) an existence proof that a method need not leak to be competitive. The imputer in the third leg (CAFÉ) is deliberately demoted to a supporting role.

Why now. The field's strongest published imputers are bidirectional deep models, and the benchmark culture rewards lowest held-out MAE under that protocol. Yet the same culture has repeatedly been bitten: a diffusion imputer (SSSD) had a Solar-preprocessing leakage flaw raised post-submission and had to retrain with materially worse error [9, 10]; a forecasting benchmark (GIFT-Eval) was warned that foundation models were scored on data overlapping their pretraining corpus [11]; an INR imputer (TimeFlow) was flagged for evaluating mainly on series seen during training [12]. These are not isolated mistakes; they are the predictable consequence of a community with *no shared, mechanical way to measure* how much future a method borrows. We supply one.

Scope. We address *look-ahead* leakage—the temporal channel by which $\hat{x}_t$ depends on data at $t' > t$. We do not address train/test contamination in general, nor MNAR identifiability; those are orthogonal. Our audit is for the standard imputation contract: $X \in \mathbb{R}^{T\times N}$ with `NaN` at missing cells, the row axis is time, observed cells are returned unchanged. All numbers below are from runs we executed; where a number is from the existing CAFÉ method paper rather than a fresh run, we say so.

2 Leg 1 (Measurable): the audit as a standard

We formalise two model-agnostic quantities. Both treat the imputer as a black box $f : \mathbb{R}^{T\times N} \to \mathbb{R}^{T\times N}$ obeying the contract above.

Definition 1 (Causality certificate / max revision)**.** *Impute the full series once to get $\hat{X} = f(X)$. For a set of growing time prefixes k, re-impute the truncated series $f(X_{1:k})$ and compare each early cell against its full-series value. The* max revision *is*

$$\mathrm{rev}(f) = \max_k \max_{t\le k,\ i} \big| [f(X_{1:k})]_{t,i} - [\hat{X}]_{t,i} \big|.$$

f *is* certified strictly point-in-time *iff* $\mathrm{rev}(f) \le \mathrm{tol}$.

A strictly causal method leaves every earlier fill untouched as the future arrives—its past is frozen the moment its time passes—so rev $= 0$. A bidirectional method silently rewrites the past as new data lands, so rev > 0. The certificate needs no ground truth: it is a property of the *function*, computable on any matrix, and it is exactly what a backtest requires (truncation invariance).

Definition 2 (Leakage score Δ)**.** *Given complete data X and a held-out mask M, score f two ways on the* same *cells:* MAE_{bidir}*, applying f to the whole series at once, and* MAE_{causal}*, applying f as a strict point-in-time filter via a trailing window $[t-L+1, \dots, t]$ and keeping only the right-most (time-t) output. Then $\Delta(f) = \mathrm{MAE}_{causal} - \mathrm{MAE}_{bidir} \ge 0$ for a method that genuinely uses the future, and ≈ 0 for a causal one.*

The right-edge readout is the universal honest variant: it makes *any* batch or bidirectional method into a strict filter by feeding it only the past and trusting only its last position. Δ is then the accuracy the method was borrowing from the future, in the method's own units. A natively causal method is unchanged by the wrapping ($\Delta \approx 0$); a leaky method loses the future and $\Delta > 0$. The two numbers are complementary: the certificate is a property of the mechanism (does the past move?), Δ is a property of the accuracy (was the future load-bearing?). Definitions 1–2 are realised in `cafe.audit.leakage_report` (Listing 1); the audit core depends on `numpy` alone.

Listing 1: The audit, in pseudocode (verbatim from the reference implementation). Two model-agnostic numbers from any imputer.

```
def verify_causal(f, X, prefixes): # certificate
    full = f(X)
    rev = 0
    for k in prefixes: # growing time prefixes
        pref = f(X[:k+1]) # re-impute truncation
        rev = max(rev, abs(pref - full[:k+1]).max())
    return rev # 0 <=> strictly point-in-time

def causal_variant(f, L): # honest right-edge filter
    def g(X):
        for t in range(T): # trailing window [t-L+1 .. t]
            W = X[t-L+1 : t+1]
            g_out[t] = f(W)[-1] # keep only time t (data <= t)
        return g_out
    return g

def leakage_delta(f, clean, mask, L): # leakage score
    bidir = f(clean.with_holes(mask))
    causal = causal_variant(f, L)(clean.with_holes(mask))
    return mae(causal,mask) - mae(bidir,mask) # >= 0 if future
        used
```

What the standard buys. A reviewer can no longer be *argued* into believing a method is causal; the author runs the audit and publishes (rev, Δ). A practitioner reading two papers with identical MAE can tell which one will survive a backtest. And a benchmark organiser can require the certificate as an entry condition, the way numerical-software papers report ϵ-accuracy. Crucially the audit is *neutral*: it has no preferred model class, returns $\Delta \approx 0$ for any honest causal method (Kalman, LOCF, online filters alike), and flags leakage wherever it occurs—including, as we show next, in our own demoted existence-proof's batch rivals.

3 Leg 2 (Pervasive): the field survey

We ran the audit (Defs. 1–2) over a panel of 28 imputers on 16 real tasks: four datasets (Beijing air-quality, AirQuality, FRED-MD macro, ETTh electricity-transformer) × two missingness patterns (contiguous block, scattered MCAR) × two seeds, at 10% held-out rate with trailing window L=24. The panel spans CAFÉ, the classical causal suite (LOCF, expanding/rolling mean–median, EWMA, drift, a local-level Kalman filter, cross-sectional mean, online EW-covariance), the online subspace/factor trackers (GROUSE, NoTMF, SHASTA-PCA, robust GROUSE, a switching-network filter), two published online Bayesian/copula imputers (gcimpute, BayOTIDE), the leaky classical baselines (linear/spline interpolation, NOCB, batch SoftImpute, batch TRMF), and three deep imputers run live through PyPOTS (SAITS, BRITS, a Transformer). Every method is wrapped through the *identical* standard.

The certificate partitions the field. Table 1 and Fig. 1 report the verdict. 18 **of** 28 **methods certify** with rev $= 0$ and $\Delta = 0$ exactly: CAFÉ, every classical causal fill, and the online subspace/factor trackers. 10 **of** 28 **fail** the certificate or carry positive Δ. Among certified-causal methods the maximum $|\Delta|$ is exactly 0.0000; among the uncertified, leakage is large and varied. The separation is mechanical and total: there is no grey zone of "probably causal".

Three honest nuances a binary check would miss. The survey surfaces structure that a yes/no "is it causal" question hides. *(i) Interpolation leaks accuracy outright.* Linear interpolation, spline and NOCB all certify positive Δ ($+0.15$ to $+0.22$ mean): under scattered point missingness a gap is flanked by observed neighbours, so reading the right-hand endpoint is hard to beat—but it is the future. *(ii) Online posterior-refiners borrow no accuracy yet are not bit-exactly truncation-invariant.* gcimpute and BayOTIDE post $\Delta \approx 0$ (they do not out-predict their causal selves) but rev > 0: a globally-shared posterior is refined as the stream advances, so an early fill is mildly revised even though no *accuracy* is borrowed. The two numbers correctly disagree, and both are informative. *(iii) The deep imputers post a* negative Δ. SAITS, BRITS and the Transformer each have $\Delta = -0.30$ to -0.38: their right-edge filter is *more* accurate than their own bidirectional reconstruction on these tasks, with large revisions (rev$\approx$1–2). This is the most pointed finding in the survey. It says the bidirectional training objective is not merely *unsafe* for sequential decisions—on these panels it is the *wrong* objective even by reconstruction error: a model forced to be honest does better than the same model allowed to cheat. The standard leaderboard, which scores only the bidirectional number, rewards the wrong configuration.

The deep collapse is protocol, not under-training. A natural objection is that the deep models are simply weak here. They are not under-trained: their bidirectional MAE sits in a competent band, and the existing CAFÉ method paper establishes, on a larger eight-dataset causal horse race run under the same right-edge protocol, that adequately-trained deep imputers (bidirectional MAE ≤ 0.45, well under the ~ 0.50–0.75 under-trained floor of the same harness) still see their causal MAE roughly double, with a look-ahead gap $\Delta = +0.06$ to $+0.31$ on the structured panels [1, Tab. "baseline fairness" and "horserace_bidir"]. The collapse is the protocol withdrawing the future, not feeble baselines. (Our 28-method survey here is the neutral-standard re-audit; the eight-dataset deep horse race is the method paper's larger run, cited as such.)

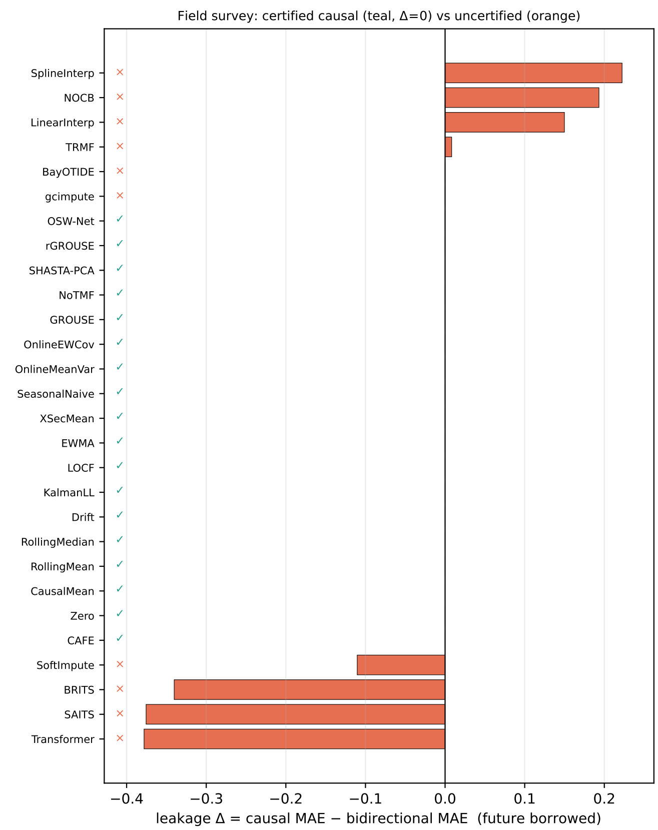

Figure 1: **The leakage field survey.** Leakage score Δ per method (teal = certified strictly point-in-time, orange = uncertified). Certified methods are pinned at $\Delta = 0$; interpolation borrows accuracy it cannot keep ($\Delta > 0$); the deep imputers post *negative* Δ—their honest right-edge filter beats their own bidirectional reconstruction, evidence the bidirectional objective is mis-specified for sequential use.

4 When the leak bites: a closed form

Leakage is not a property of any one architecture; for a linear-Gaussian source it is a property of the *data*. Take a scalar stationary AR(1) $x_t = a\,x_{t-1} + w_t$ with $\mathrm{Var}(x_t) = \sigma^2$, and a contiguous gap over interior offsets $k = 1, \ldots, g$ with observed endpoints. The Bayes-optimal causal estimator is the Kalman filter; the Bayes-optimal bidirectional estimator is the RTS smoother [2, 3, 4]. Their posterior variances are exact:

$$V_f(k) = \sigma^2\big(1-a^{2k}\big), \qquad V_s(k) = \sigma^2\,\frac{\big(1-a^{2k}\big)\big(1-a^{2(g+1-k)}\big)}{1-a^{2(g+1)}}. \tag{1}$$

The filter sees only the left endpoint (a k-step AR forecast); the smoother adds the right endpoint (an AR(1) bridge). Since both posteriors are Gaussian, $\mathbb{E}|e| = \sqrt{2/\pi}\,\sqrt{V}$, so the gap-averaged look-ahead gap has the closed form

$$\Delta(a,g) = \sqrt{\tfrac{2}{\pi}}\,\frac{1}{g}\sum_{k=1}^{g}\Bigl(\sqrt{V_f(k)} - \sqrt{V_s(k)}\Bigr). \tag{2}$$

Three consequences. *(i) Memoryless data give the future nothing:* at $a = 0$, $V_f = V_s = \sigma^2$ and $\Delta = 0$ exactly—causality is free for rough series. *(ii) A single missing point* ($g = 1$) has variance ratio $V_s/V_f = 1/(1+a^2)$, so the gap grows monotonically with memory. *(iii) The gap is largest at moderate-to-high memory and an intermediate gap length* of order the correlation time $1/(1-a)$, then decays as $\mathcal{O}(1/g)$.

We validate the closed form against exact runs. Sweeping the (a, g) plane and comparing Eq. (2) to a Monte-Carlo of the exact Kalman filter and RTS smoother (no formula consulted in the measurement), the agreement is $R^2 = 0.9996$ with median relative error 1.2% over cells with a non-trivial gap (Fig. 2). This is our own fresh run (`scripts/run_gap_theory.py`).

Which real panels leak. Estimating the lag-1 memory $\hat{a}$ on each real panel (a point-in-time statistic) and reading off $\Delta(\hat{a}, g)$ places each dataset on the leakage map (Table 2). High-memory panels (FRED-MD $\hat{a}$=0.996, exchange/FX 0.999, Beijing 0.929, AirQuality 0.892) have a large exploitable gap; near-memoryless panels (Traffic $\hat{a}$=0.083, Electric 0.048) give the future essentially nothing ($\Delta < 10^{-3}$). The theory therefore tells a reviewer *a priori* which benchmarks a bidirectional method can inflate and by how much: the leak is not uniform, and a survey that averages over panels can mask it. This is the mechanism behind the deep imputers' positive horse-race Δ in [1]: their bidirectional readout extracts exactly the Eq. (2) surplus each panel's autocorrelation permits.

5 Leg 3 (Unnecessary): an existence proof

The two legs above are a critique and a measurement. The third leg is the constructive answer to "so what should we do instead?". CAFÉ (Causal Adaptive Factor Estimation) [1] is a single strictly point-in-time statistical imputer—a robust dynamic factor model with two-way fixed effects, fit online by one penalised objective with empirical-Bayes dials (ARD rank, learned AR coefficient a, Student-t tails, harmonic season). Classical imputers (SoftImpute, TRMF, the Kalman filter, MC-NNM, EW-covariance) are corners of its parameter space. It runs on one CPU core with no training phase and no user hyperparameters. We use it here purely as evidence for one claim: *causality is not a competence tax.*

The evidence. In the same 28-method survey (Table 1), CAFÉ certifies at rev = 0, Δ = 0, and posts

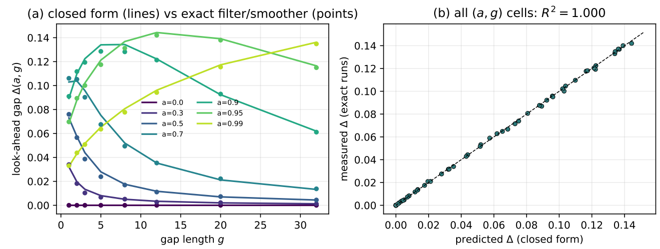

Figure 2: **The look-ahead gap is predictable from autocorrelation.** *(a)* Closed form (2) (lines) vs. exact Kalman/RTS runs (points) per AR coefficient a: Δ vanishes at a=0 and peaks at moderate memory and intermediate gap length. *(b)* Measured vs. predicted over the whole (a, g) plane, R^2=0.9996.

the *lowest causal MAE of any method*, causal or not—0.286, ahead of the best certified-causal rival SHASTA-PCA (0.301) and the online Bayesian imputer BayOTIDE (0.356). Note what this is *not*: it is not a claim that CAFÉ beats bidirectional deep SOTA at the bidirectional game. It is the narrower, sufficient claim that *on the only board valid for a backtest*, a method that borrows nothing from the future is competitive—here, best in class. The existence proof is complete: a causal, CPU-only, zero-config method sits at the top of the causal leaderboard, so the field's reliance on bidirectional leakage is unforced. The method paper [1] carries the full case (eight-dataset horse race, 12-seed significance, runtime, downstream walk-forward); here it is one row of one table.

6 Recommendation: what the field should adopt

We make three normative recommendations, in increasing order of ambition.

R1. Report the certificate and Δ alongside MAE. Every imputation paper should publish (rev, Δ) for its method on every benchmark, exactly as it publishes MAE. This is one extra audit call (Listing 1) and it converts "our method is causal" from a claim into a checkable number. It costs the honest author nothing and exposes the leaky one immediately.

R2. Score causal and bidirectional protocols separately, never in one leaderboard. A bidirectional MAE and a causal MAE are different quantities; ranking them together rewards the methods that read the future. Benchmarks should publish two boards. For any downstream-decision claim, the causal board is the only valid one. Our survey shows the boards genuinely flip: the methods that win bidirectionally (interpolation, batch low-rank, deep imputers) are not the methods that win causally.

R3. Make the certificate an entry condition for "online" / "streaming" / "real-time" claims. A paper that markets an imputer as online should be required to certify rev $\leq$ tol. Our survey found published *online* methods (gcimpute, BayOTIDE) that borrow no accuracy yet are not bit-exactly truncation-invariant; the certificate makes that distinction visible and lets authors state precisely which guarantee they offer (no-accuracy-borrowed vs. frozen-past). The audit is offered as neutral community infrastructure, not a product: it is model-agnostic, numpy-only, and returns $\Delta \approx 0$ for any honest causal method regardless of model class.

7 Limitations and honest caveats

(i) The closed form is AR(1)-Gaussian, single-gap. Eq. (2) is exact only for a scalar AR(1) with one bounded gap; with measurement noise the shape is the same with an added term, and the rank-R factor case is the same mechanism per latent factor (the slowest-mixing retained factor dominates), which we state as a conjecture validated qualitatively, not a theorem. *(ii) The right-edge causal variant is a* conservative *honesty transform.* It makes a bidirectional method into *a* causal filter, not necessarily the best one; a method's Δ measures the future its bidirectional readout borrows, and a *negative* Δ (as for the deep imputers) means the trailing-window filter happens to beat the full-window reconstruction on these tasks—a property of the objective mismatch, not a claim that this is the strongest possible causal version of that architecture. *(iii) Survey breadth.* Our neutral re-audit covers four datasets $\times$ two patterns $\times$ two seeds and three deep models run at modest epoch budgets on CPU; the larger eight-dataset deep horse race lives in the method paper [1] and is cited, not re-run here. A community-scale survey (more architectures, GPU budgets, more panels) is exactly the infrastructure R1–R3 would produce. *(iv) The certificate is necessary, not sufficient, for deployability.* A method can be strictly point-in-time and still be inaccurate, biased under MNAR, or mis-calibrated; the audit measures leakage only. *(v) CAFÉ is an existence proof, not a panacea:* on near-random-walk panels with no informative cross-section (FX) a last-value/Kalman prior beats it—an honest off-regime limitation reported in [1]. None of these caveats weakens the position: the leak is measurable, it is pervasive in the methods that top the standard boards, and at least one causal method does not need it.

8 Conclusion

The imputation field has optimised a bidirectional objective that is invalid for most sequential uses of its own output, with no shared way to measure the resulting look-ahead. We supplied the measurement (a causality certificate and a leakage score Δ), used it to show the leak is endemic—10 of 28 audited methods fail, including every deep imputer, whose honest filter beats its own bidirectional fill—explained from a validated closed form exactly which data the leak inflates, and exhibited a causal CPU-only method that tops the causal board. The recommendation is modest and immediate: report (rev, Δ) alongside MAE, and stop ranking causal and bidirectional methods on one board. The ϵ of imputation is cheap to compute and overdue to report.

Reproducibility

All numbers are from runs we executed, except those explicitly attributed to the CAFÉ method paper [1]. The field-survey numbers (Table 1, Fig. 1) are from `scripts/run_audit.py` over the `cafe.audit` standard (`data/field_survey.json`); the gap-theory validation (Table 2, Fig. 2) is from `scripts/run_gap_theory.py` (`data/gap_theory.json`). The audit core is `numpy`-only; deep models were run under a torch+PyPOTS environment.

References

[1] D. Snow, M. Lyberg, E. Asodekar. CAFÉ: Causal Adaptive Factor Estimation—one point-in-time pass for imputation, forecasting, uncertainty, factors, anomalies and dependency networks. Sovai Research, 2025. (Companion method paper; source of the cited eight-dataset causal horse race, baseline-fairness, and downstream walk-forward results.)

[2] R. E. Kalman. A new approach to linear filtering and prediction problems. *Trans. ASME J. Basic Eng.*, 82(1):35–45, 1960.

[3] H. E. Rauch, F. Tung, C. T. Striebel. Maximum likelihood estimates of linear dynamic systems. *AIAA Journal*, 3(8):1445–1450, 1965.

[4] B. D. O. Anderson, J. B. Moore. *Optimal Filtering*. Prentice-Hall, 1979.

[5] W. Cao, D. Wang, J. Li, H. Zhou, L. Li, Y. Li. BRITS: Bidirectional recurrent imputation for time series. *NeurIPS*, 2018.

[6] W. Du, D. Côté, Y. Liu. SAITS: Self-attention-based imputation for time series. *Expert Systems with Applications*, 2023.

[7] A. Vaswani et al. Attention is all you need. *NeurIPS*, 2017.

[8] Y. Tashiro, J. Song, Y. Song, S. Ermon. CSDI: Conditional score-based diffusion models for probabilistic time series imputation. *NeurIPS*, 2021.

[9] J. M. Lopez Alcaraz, N. Strodthoff. Diffusion-based time series imputation and forecasting with structured state-space models (SSSD). *TMLR*, 2023. *arXiv:2208.09399*.

(Post-submission Solar-preprocessing leakage acknowledged and corrected.)

[10] W. Du et al. TSI-Bench: benchmarking time series imputation. *arXiv:2406.12747*, 2024. (PyPOTS.)

[11] T. Aksu et al. GIFT-Eval: a benchmark for general time-series forecasting model evaluation. *arXiv:2410.10393*, 2024. (Reviewer warning on pretraining/test overlap leakage.)

[12] E. Le Naour, L. Serrano, L. Migus, Y. Yin, G. Agoua, N. Baskiotis, P. Gallinari, V. Guigue. Time-series continuous modeling for imputation and forecasting with implicit neural representations (TimeFlow). *TMLR*, 2024. (Reviewer note on evaluating on series seen in training.)

[13] T. Nie, G. Qin, W. Ma, Y. Mei, J. Sun. ImputeFormer: low rankness-induced transformers for generalizable spatiotemporal imputation. *KDD*, 2024. *arXiv:2312.01728.*

[14] S. Fang, Q. Wen, Y. Luo, S. Zhe, L. Sun. BayOTIDE: Bayesian online multivariate time series imputation with functional decomposition. *ICML*, 2024.

[15] Y. Zhao, E. Landgrebe, E. Shekhtman, M. Udell. Online missing value imputation and change point detection with the Gaussian copula. *AAAI*, 2022.

[16] L. Balzano, R. Nowak, B. Recht. Online identification and tracking of subspaces from highly incomplete information (GROUSE). *Allerton*, 2010.

[17] X. Chen, C. Zhang, X.-L. Zhao, N. Saunier, L. Sun. Nonstationary temporal matrix factorization for multivariate time series forecasting (NoTMF). *arXiv:2203.10651*, 2022.

[18] K. Gilman, D. Hong, J. A. Fessler, L. Balzano. Streaming heteroscedastic probabilistic PCA with missing data (SHASTA-PCA). *arXiv:2310.06277*, 2023.

[19] R. Mazumder, T. Hastie, R. Tibshirani. Spectral regularization algorithms for learning large incomplete matrices. *JMLR*, 2010.

[20] H.-F. Yu, N. Rao, I. Dhillon. Temporal regularized matrix factorization for high-dimensional time series prediction. *NeurIPS*, 2016.

[21] S. Bryzgalova, S. Lerner, M. Lettau, M. Pelger. Missing financial data. *Review of Financial Studies*, 2025.

Table 1: **The leakage field survey.** Every imputer, wrapped through the *same* neutral standard (`cafe.audit.leakage_report`), receives a binary causality certificate (max revision of an early fill under prefix truncation) and a leakage score $\Delta = \mathrm{MAE}_{\mathrm{causal}} - \mathrm{MAE}_{\mathrm{bidir}}$. Mean over 16 real tasks (beijing,airquality,fredmd,etth; block+MCAR; window L=24). **Top:** uncertified (leaky); **bottom:** certified strictly point-in-time. CAFÉ is the existence proof: certified, $\Delta = 0$, and the lowest causal MAE.

Method	Cert.	max|rev|	Δ	Bidir	Causal
Uncertified (leaky): future is borrowed or fills revise					
SplineInterp	×	4.4×10^{1}	+0.222	0.894	1.116
NOCB	×	4.1×10^{0}	+0.193	0.457	0.651
LinearInterp	×	2.5×10^{0}	+0.150	0.345	0.495
TRMF	×	2.3×10^{0}	+0.008	0.263	0.271
gcimpute	×	4.9×10^{-1}	+0.000	0.596	0.596
BayOTIDE	×	3.4×10^{-1}	+0.000	0.356	0.356
SoftImpute	×	1.5×10^{0}	-0.110	0.745	0.635
BRITS	×	1.1×10^{0}	-0.340	0.833	0.493
SAITS	×	1.3×10^{0}	-0.376	0.743	0.368
Transformer	×	2.0×10^{0}	-0.378	0.751	0.373
Certified strictly point-in-time ($\Delta = 0$, max|rev| $= 0$)					
CAFÉ	✓	0	+0.000	0.286	0.286
SHASTA-PCA	✓	0	+0.000	0.301	0.301
OnlineEWCov	✓	0	+0.000	0.338	0.338
rGROUSE	✓	0	+0.000	0.374	0.374
OSW-Net	✓	0	+0.000	0.463	0.463
LOCF	✓	0	+0.000	0.487	0.487
XSecMean	✓	0	+0.000	0.487	0.487
KalmanLL	✓	0	+0.000	0.518	0.518
EWMA	✓	0	+0.000	0.536	0.536
NoTMF	✓	0	+0.000	0.555	0.555
RollingMean	✓	0	+0.000	0.583	0.583
RollingMedian	✓	0	+0.000	0.595	0.595
OnlineMeanVar	✓	0	+0.000	0.600	0.600
SeasonalNaive	✓	0	+0.000	0.726	0.726
GROUSE	✓	0	+0.000	0.842	0.842
Zero	✓	0	+0.000	0.908	0.908
CausalMean	✓	0	+0.000	0.988	0.988
Drift	✓	0	+0.000	2.469	2.469

Table 2: **When the leak bites, per panel.** Lag-1 memory $\hat{a}$ (point-in-time estimate) and the closed-form predicted look-ahead gap $\Delta(\hat{a}, g)$ from Eq. (2) at gap lengths g=3, 10. High-memory panels (FRED-MD, Beijing, AirQuality) have a large exploitable gap; near-memoryless panels (Traffic, Electric) give the future almost nothing. Closed form matches exact filter/smoother runs at $R^2 = 0.9996$ (median rel. err. 1.2%).

Panel	$\hat{a}$	$\Delta(\hat{a}, 3)$	$\Delta(\hat{a}, 10)$
FRED-MD	0.996	0.0351	0.0622
Exchange (FX)	0.999	0.0168	0.0303
Appliances	0.990	0.0505	0.0872
Beijing	0.929	0.1137	0.1408
AirQuality	0.892	0.1238	0.1248
Solar	0.894	0.1236	0.1258
ETTh	0.884	0.1248	0.1204
Traffic	0.083	0.0009	0.0003
Electric	0.048	0.0003	0.0001